

VERIFICATION & RESULTS

Evaluation Report

All twelve assignment cases run end to end, twice — once as structured data and once as real document uploads — with the decision, the reasons, and the complete step-by-step trace for each.

Generated by `uv run python -m app.eval.runner` · August 11, 2026

RESULTS AT A GLANCE

STRUCTURED RUN

12 / 12

deterministic — the canonical verdict

UPLOAD RUN

12 / 12

real documents through GPT-4o

PER DECISION

Full trace

every step, rule reference and delta

12/12 cases matched expected outcomes.

Generated by `uv run python -m app.eval.runner` (from `backend/`). Every case shows the full decision output and the complete trace.

The same 12 also run as **real document uploads** through GPT-4o classification and extraction: **12/12 matched** (`--with-uploads`). Two paths, one set of expectations — a decision that only holds when documents arrive pre-typed has not been demonstrated. The structured run below is the canonical verdict because it is deterministic; the upload run is what a member actually triggers, and is the only path that can catch a fault in *reading* a document.

CASE	NAME	EXPECTED	STRUCTURED	UPLOAD	MATCH
TC001	Wrong Document Uploaded	— (stop)	— (stopped: document problems) ✓	— (stopped: document problems) ✓	✓
TC002	Unreadable Document	— (stop)	— (stopped: document problems) ✓	— (stopped: document problems) ✓	✓
TC003	Documents Belong to Different Patients	— (stop)	— (stopped: document problems) ✓	— (stopped: document problems) ✓	✓
TC004	Clean Consultation — Full Approval	APPROVED	APPROVED ✓	APPROVED ✓	✓
TC005	Waiting Period — Diabetes	REJECTED	REJECTED ✓	REJECTED ✓	✓
TC006	Dental Partial Approval — Cosmetic Exclusion	PARTIAL	PARTIAL ✓	PARTIAL ✓	✓
TC007	MRI Without Pre-Authorization	REJECTED	REJECTED ✓	REJECTED ✓	✓
TC008	Per-Claim Limit Exceeded	REJECTED	REJECTED ✓	REJECTED ✓	✓
TC009	Fraud Signal — Multiple Same-Day Claims	MANUAL_REVIEW	MANUAL_REVIEW ✓	MANUAL_REVIEW ✓	✓
TC010	Network Hospital — Discount Applied	APPROVED	APPROVED ✓	APPROVED ✓	✓
TC011	Component Failure — Graceful Degradation	APPROVED	APPROVED ✓	APPROVED ✓	✓
TC012	Excluded Treatment	REJECTED	REJECTED ✓	REJECTED ✓	✓

TC001 Wrong Document Uploaded ✓

Expected: early stop, no decision

Got: — (stopped: document problems)

Member-facing problems:

- **[WRONG_TYPE]** A CONSULTATION claim requires: PRESCRIPTION and HOSPITAL_BILL. You uploaded PRESCRIPTION (dr_sharma_prescription.jpg), PRESCRIPTION (another_prescription.jpg) — a PRESCRIPTION was provided where a HOSPITAL_BILL is required, and no HOSPITAL_BILL was included.
 - *Action:* Upload the hospital bill for this treatment and resubmit the claim.

system_must checks:

- ✓ names uploaded and required types

FULL TRACE — 3 STEPS

#	COMPONENT	ACTION	OUTCOME	TOOK	Δ CONF	DETAIL
1	intake_validator	validate submission payload	PASS	<1ms	—	Claim by EMP001 for CONSULTATION, treatment 2024-11-01, amount ₹1500, 2 document(s).
2	document_verifier	required document types	FAIL	<1ms	—	Missing required type(s): HOSPITAL_BILL. Uploaded: PRESCRIPTION (dr_sharma_prescription.jpg), PRESCRIPTION (another_prescription.jpg).
3	document_verifier	verification stopped	FAIL	<1ms	—	1 document problem(s) found; processing stopped before any decision. The claim is returned to the member with instructions.

TC002 Unreadable Document ✓

Expected: early stop, no decision
Got: — (stopped: document problems)

Member-facing problems:

- [UNREADABLE] 'blurry_bill.jpg' is too blurred to read — no text could be recovered from it, so we cannot tell what document it is or what it says. Your claim has not been rejected and nothing else is wrong with it. Your other document is fine — only this one needs replacing.
 - Action: Re-upload 'blurry_bill.jpg': photograph the original in good light with the whole page flat in frame, hold steady until the camera focuses, and check the text is readable before submitting. A scanned PDF works too.

system_must checks:

- ✓ asks re-upload of the specific document

FULL TRACE — 4 STEPS

#	COMPONENT	ACTION	OUTCOME	TOOK	Δ CONF	DETAIL
1	intake_validator	validate submission payload	PASS	<1ms	—	Claim by EMP004 for PHARMACY, treatment 2024-10-25, amount ₹800, 2 document(s).
2	document_verifier	required document types	PASS	<1ms	—	All required types for PHARMACY present (PRESCRIPTION, PHARMACY_BILL). Uploaded: PRESCRIPTION (prescription.jpg), PHARMACY_BILL (blurry_bill.jpg).
3	document_verifier	readability check	FAIL	<1ms	—	'blurry_bill.jpg' is unreadable. Claim not rejected; a re-upload of this one document is all that is required.
4	document_verifier	verification stopped	FAIL	<1ms	—	1 document problem(s) found; processing stopped before any decision. The claim is returned to the member with instructions.

TC003

Documents Belong to Different Patients ✓

Expected: early stop, no decision
Got: — (stopped: document problems)

Member-facing problems:

- [PATIENT_MISMATCH]** The uploaded documents belong to different patients: PRESCRIPTION ('prescription_rajesh.jpg') is for Rajesh Kumar; HOSPITAL_BILL ('bill_arjun.jpg') is for Arjun Mehta. All documents in one claim must be for the same person.
 - Action:* Check the files and re-upload documents that all belong to the patient this claim is for. If you are claiming for a dependent, every document must carry the dependent's name.

system_must checks:

- ✓ names both patients

FULL TRACE — 5 STEPS

#	COMPONENT	ACTION	OUTCOME	TOOK	Δ CONF	DETAIL
1	intake_validator	validate submission payload	PASS	<1ms	—	Claim by EMP001 for CONSULTATION, treatment 2024-11-01, amount ₹1500, 2 document(s).
2	document_verifier	required document types	PASS	<1ms	—	All required types for CONSULTATION present (PRESCRIPTION, HOSPITAL_BILL). Uploaded: PRESCRIPTION (prescription_rajesh.jpg), HOSPITAL_BILL (bill_arjun.jpg).
3	extraction_agent	extract structured data from documents	SKIPPED	<1ms	—	Pre-extracted content supplied with the submission; vision extraction skipped (test mode).
4	document_verifier	patient consistency	FAIL	<1ms	—	Documents name different patients: PRESCRIPTION ('prescription_rajesh.jpg') is for Rajesh Kumar; HOSPITAL_BILL ('bill_arjun.jpg') is for Arjun Mehta.
5	document_verifier	verification stopped	FAIL	<1ms	—	1 document problem(s) found after reading the documents; processing stopped before any decision.

TC004

Clean Consultation — Full Approval ✓

Expected: APPROVED, ₹1,350
Got: APPROVED

Approved: ₹1,350 · **Confidence:** 0.98

Reasons given:

- Approved ₹1,350 of claimed ₹1,500.
- Co-pay (10%) borne by member. -₹150 (₹1,500 → ₹1,350)

Line items:

ITEM	CLAIMED	VERDICT	REASON
Consultation Fee	₹1,000	✓ approved	—
CBC Test	₹300	✓ approved	—
Dengue NS1 Test	₹200	✓ approved	—

FULL TRACE — 20 STEPS

#	COMPONENT	ACTION	OUTCOME	TOOK	Δ CONF	DETAIL
1	intake_validator	validate submission payload	PASS	<1ms	—	Claim by EMP001 for CONSULTATION, treatment 2024-11-01, amount ₹1500, 2 document(s).
2	document_verifier	required document types	PASS	<1ms	—	All required types for CONSULTATION present (PRESCRIPTION, HOSPITAL_BILL). Uploaded: PRESCRIPTION, HOSPITAL_BILL.
3	extraction_agent	extract structured data from documents	SKIPPED	<1ms	—	Pre-extracted content supplied with the submission; vision extraction skipped (test mode).
4	document_verifier	patient consistency	PASS	<1ms	—	All named documents agree on patient 'Rajesh Kumar'.
5	policy_retriever	rule source	PASS	<1ms	—	In-memory policy snapshot (authoritative; no policy graph configured).
6	adjudication_engine	eligibility.member	PASS	1ms	—	Member EMP001 (Rajesh Kumar, SELF) found in roster.
7	adjudication_engine	eligibility.policy	PASS	<1ms	—	Policy PLUM_GHI_2024 active on treatment date 2024-11-01.
8	adjudication_engine	submission.deadline	PASS	<1ms	—	Submitted 0 days after treatment, within the 30-day deadline.
9	adjudication_engine	submission.minimum_amount	PASS	<1ms	—	Claimed amount ₹1,500 meets the ₹500 minimum.
10	adjudication_engine	amount_reconciliation	PASS	<1ms	—	Claimed ₹1,500 matches the documented ₹1,500 (itemized line items).
11	adjudication_engine	exclusions.claim_level	PASS	<1ms	—	Diagnosis/treatment ('Viral Fever') matches no policy exclusion.
12	adjudication_engine	waiting_period.initial	PASS	<1ms	—	Initial 30-day waiting period complete (joined 2024-04-01).
13	adjudication_engine	waiting_period.condition_specific	PASS	<1ms	—	Diagnosis 'Viral Fever' matches no condition-specific waiting period.
14	adjudication_engine	pre_authorization	PASS	<1ms	—	No pre-authorization rules apply to CONSULTATION claims.
15	adjudication_engine	line_items	PASS	<1ms	—	All 3 line item(s) covered under consultation.
16	adjudication_engine	per_claim_limit	PASS	<1ms	—	Eligible amount ₹1,500 is within the ₹5,000 per-claim limit.

17	adjudication_engine	financial computation	PASS	<1ms	—	eligible_base ₹1500.00 → network_discount ₹1500.00 → copay ₹1350.00 → sub_limit_cap ₹1350.00 → annual_limit_cap ₹1350.00
18	fraud_checker	velocity and threshold rules	PASS	<1ms	—	No fraud signals: same-day, monthly, and high-value checks all within limits.
19	decision_synthesizer	confidence rollup	PASS	<1ms	—	Confidence 0.98 = base 0.98; no penalties.
20	decision_synthesizer	final decision	PASS	<1ms	—	APPROVED; approved amount ₹1,350; confidence 0.98.

TC005

Waiting Period — Diabetes ✓

Expected: REJECTED, reasons ['WAITING_PERIOD']

Got: REJECTED

Approved: ₹0 · **Confidence:** 0.93

Reasons given:

- Diagnosis 'Type 2 Diabetes Mellitus' falls under the 90-day waiting period for diabetes (member joined 2024-09-01). Treatment on 2024-10-15 is inside this period. The member is eligible for diabetes-related claims from 2024-11-30.
- Confidence 0.93 (from 0.98); the following were noted.
- bill not itemized; adjudicated on billed total
- Bill has no itemized line items; adjudicated against the billed total.

Line items:

ITEM	CLAIMED	VERDICT	REASON
Billed amount (no itemized bill)	₹3,000	✓ approved	—

system_must checks:

- ✓ states the eligibility date

FULL TRACE — 19 STEPS

#	COMPONENT	ACTION	OUTCOME	TOOK	Δ CONF	DETAIL
1	intake_validator	validate submission payload	PASS	<1ms	—	Claim by EMP005 for CONSULTATION, treatment 2024-10-15, amount ₹3000, 2 document(s).
2	document_verifier	required document types	PASS	<1ms	—	All required types for CONSULTATION present (PRESCRIPTION, HOSPITAL_BILL). Uploaded: PRESCRIPTION, HOSPITAL_BILL.
3	extraction_agent	extract structured data from documents	SKIPPED	<1ms	—	Pre-extracted content supplied with the submission; vision extraction skipped (test mode).
4	document_verifier	patient consistency	PASS	<1ms	—	All named documents agree on patient 'Vikram Joshi'.
5	policy_retriever	rule source	PASS	<1ms	—	In-memory policy snapshot (authoritative; no policy graph configured).

6	adjudication_engine	eligibility.member	PASS	<1ms	—	Member EMP005 (Vikram Joshi, SELF) found in roster.
7	adjudication_engine	eligibility.policy	PASS	<1ms	—	Policy PLUM_GHI_2024 active on treatment date 2024-10-15.
8	adjudication_engine	submission.deadline	PASS	<1ms	—	Submitted 0 days after treatment, within the 30-day deadline.
9	adjudication_engine	submission.minimum_amount	PASS	<1ms	—	Claimed amount ₹3,000 meets the ₹500 minimum.
10	adjudication_engine	amount_reconciliation	PASS	<1ms	—	Claimed ₹3,000 matches the documented ₹3,000 (billed total).
11	adjudication_engine	exclusions.claim_level	PASS	<1ms	—	Diagnosis/treatment ('Type 2 Diabetes Mellitus') matches no policy exclusion.
12	adjudication_engine	waiting_period.initial	PASS	<1ms	—	Initial 30-day waiting period complete (joined 2024-09-01).
13	adjudication_engine	waiting_period.diabetes	FAIL	<1ms	—	Diagnosis 'Type 2 Diabetes Mellitus' falls under the 90-day waiting period for diabetes (member joined 2024-09-01). Treatment on 2024-10-15 is inside this period. The member is eligible for diabetes-related claims from 2024-11-30.
14	adjudication_engine	pre_authorization	PASS	<1ms	—	No pre-authorization rules apply to CONSULTATION claims.
15	adjudication_engine	line_items	PASS	<1ms	—	All 1 line item(s) covered under consultation.
16	adjudication_engine	per_claim_limit	PASS	<1ms	—	Eligible amount ₹3,000 is within the ₹5,000 per-claim limit.
17	fraud_checker	velocity and threshold rules	PASS	<1ms	—	No fraud signals: same-day, monthly, and high-value checks all within limits.
18	decision_synthesizer	confidence rollup	PASS	<1ms	—	Confidence 0.93 = base 0.98 -0.05 (bill not itemized; adjudicated on billed total)
19	decision_synthesizer	final decision	FAIL	<1ms	—	REJECTED; approved amount ₹0; confidence 0.93.

TC006

Dental Partial Approval — Cosmetic Exclusion ✓

Expected: PARTIAL, ₹8,000

Got: PARTIAL

Approved: ₹8,000 · **Confidence:** 0.98

Reasons given:

- Partially approved ₹8,000 of claimed ₹12,000.
- Line item rejected — 'Teeth Whitening' is an excluded procedure under the dental category ('Teeth Whitening').
- YTD claims amount not provided; assumed ₹0 for the annual OPD limit check.

Line items:

ITEM	CLAIMED	VERDICT	REASON
Root Canal Treatment	₹8,000	✓ approved	—

Teeth Whitening	₹4,000	✖ rejected	'Teeth Whitening' is an excluded procedure under the dental category ('Teeth Whitening').
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system_must checks:

- ✓ itemizes line-item verdicts

FULL TRACE — 19 STEPS

#	COMPONENT	ACTION	OUTCOME	TOOK	Δ CONF	DETAIL
1	intake_validator	validate submission payload	PASS	<1ms	—	Claim by EMPO02 for DENTAL, treatment 2024-10-15, amount ₹12000, 1 document(s).
2	document_verifier	required document types	PASS	<1ms	—	All required types for DENTAL present (HOSPITAL_BILL). Uploaded: HOSPITAL_BILL.
3	extraction_agent	extract structured data from documents	SKIPPED	<1ms	—	Pre-extracted content supplied with the submission; vision extraction skipped (test mode).
4	document_verifier	patient consistency	PASS	<1ms	—	All named documents agree on patient 'Priya Singh'.
5	policy_retriever	rule source	PASS	<1ms	—	In-memory policy snapshot (authoritative; no policy graph configured).
6	adjudication_engine	eligibility.member	PASS	4ms	—	Member EMPO02 (Priya Singh, SELF) found in roster.
7	adjudication_engine	eligibility.policy	PASS	<1ms	—	Policy PLUM_GHI_2024 active on treatment date 2024-10-15.
8	adjudication_engine	submission.deadline	PASS	<1ms	—	Submitted 0 days after treatment, within the 30-day deadline.
9	adjudication_engine	submission.minimum_amount	PASS	<1ms	—	Claimed amount ₹12,000 meets the ₹500 minimum.
10	adjudication_engine	amount_reconciliation	PASS	<1ms	—	Claimed ₹12,000 matches the documented ₹12,000 (itemized line items).
11	adjudication_engine	exclusions.claim_level	SKIPPED	<1ms	—	No diagnosis or treatment text available to screen against exclusions.
12	adjudication_engine	waiting_period.initial	PASS	<1ms	—	Initial 30-day waiting period complete (joined 2024-04-01).
13	adjudication_engine	pre_authorization	PASS	<1ms	—	No pre-authorization rules apply to DENTAL claims.
14	adjudication_engine	line_items	FAIL	<1ms	—	1 of 2 line item(s) approved; 'Teeth Whitening' rejected: 'Teeth Whitening' is an excluded procedure under the dental category ('Teeth Whitening').
15	adjudication_engine	per_claim_limit	PASS	<1ms	—	Eligible amount ₹8,000 is within the ₹10,000 per-claim limit.

16	adjudication_engine	financial computation	PASS	<1ms	—	eligible_base ₹8000.00 → network_discount ₹8000.00 → copay ₹8000.00 → sub_limit_cap ₹8000.00 → annual_limit_cap ₹8000.00
17	fraud_checker	velocity and threshold rules	PASS	<1ms	—	No fraud signals: same-day, monthly, and high-value checks all within limits.
18	decision_synthesizer	confidence rollup	PASS	<1ms	—	Confidence 0.98 = base 0.98; no penalties.
19	decision_synthesizer	final decision	PASS	<1ms	—	PARTIAL; approved amount ₹8,000; confidence 0.98.

TC007

MRI Without Pre-Authorization ✓

Expected: REJECTED, reasons ['PRE_AUTH_MISSING']

Got: REJECTED

Approved: ₹0 · **Confidence:** 0.98

Reasons given:

- 'MRI Lumbar Spine' (₹15,000) requires pre-authorization because MRI above ₹10,000 must be pre-approved, and no authorization was obtained. To resubmit: request pre-authorization from the insurer for this procedure and attach the approval reference (valid 30 days from issue).
- Also failed: Claimed amount ₹15,000 (eligible ₹15,000) exceeds the per-claim limit of ₹10,000 for DIAGNOSTIC claims. Claims above this limit are not payable.

Line items:

ITEM	CLAIMED	VERDICT	REASON
MRI Lumbar Spine	₹15,000	✓ approved	—

system_must checks:

- ✓ explains pre-auth and resubmission

FULL TRACE — 19 STEPS

#	COMPONENT	ACTION	OUTCOME	TOOK	Δ CONF	DETAIL
1	intake_validator	validate submission payload	PASS	<1ms	—	Claim by EMPO07 for DIAGNOSTIC, treatment 2024-11-02, amount ₹15000, 3 document(s).
2	document_verifier	required document types	PASS	<1ms	—	All required types for DIAGNOSTIC present (PRESCRIPTION, LAB_REPORT, HOSPITAL_BILL). Uploaded: PRESCRIPTION, LAB_REPORT, HOSPITAL_BILL.
3	extraction_agent	extract structured data from documents	SKIPPED	<1ms	—	Pre-extracted content supplied with the submission; vision extraction skipped (test mode).
4	document_verifier	patient consistency	SKIPPED	<1ms	—	No patient name could be read from any document; nothing to cross-check.
5	policy_retriever	rule source	PASS	<1ms	—	In-memory policy snapshot (authoritative; no policy graph configured).
6	adjudication_engine	eligibility.member	PASS	3ms	—	Member EMPO07 (Suresh Patil, SELF) found in roster.

7	adjudication_engine	eligibility.policy	PASS	<1ms	—	Policy PLUM_GHI_2024 active on treatment date 2024-11-02.
8	adjudication_engine	submission.deadline	PASS	<1ms	—	Submitted 0 days after treatment, within the 30-day deadline.
9	adjudication_engine	submission.minimum_amount	PASS	<1ms	—	Claimed amount ₹15,000 meets the ₹500 minimum.
10	adjudication_engine	amount_reconciliation	PASS	<1ms	—	Claimed ₹15,000 matches the documented ₹15,000 (itemized line items).
11	adjudication_engine	exclusions.claim_level	PASS	<1ms	—	Diagnosis/treatment ('Suspected Lumbar Disc Herniation') matches no policy exclusion.
12	adjudication_engine	waiting_period.initial	PASS	<1ms	—	Initial 30-day waiting period complete (joined 2024-04-01).
13	adjudication_engine	waiting_period.condition_specific	PASS	<1ms	—	Diagnosis 'Suspected Lumbar Disc Herniation' matches no condition-specific waiting period.
14	adjudication_engine	pre_authorization	FAIL	<1ms	—	'MRI Lumbar Spine' (₹15,000) requires pre-authorization because MRI above ₹10,000 must be pre-approved, and no authorization was obtained. To resubmit: request pre-authorization from the insurer for this procedure and attach the approval reference (valid 30 days from issue).
15	adjudication_engine	line_items	PASS	<1ms	—	All 1 line item(s) covered under diagnostic.
16	adjudication_engine	per_claim_limit	FAIL	<1ms	—	Claimed amount ₹15,000 (eligible ₹15,000) exceeds the per-claim limit of ₹10,000 for DIAGNOSTIC claims. Claims above this limit are not payable.
17	fraud_checker	velocity and threshold rules	PASS	<1ms	—	No fraud signals: same-day, monthly, and high-value checks all within limits.
18	decision_synthesizer	confidence rollup	PASS	<1ms	—	Confidence 0.98 = base 0.98; no penalties.
19	decision_synthesizer	final decision	FAIL	<1ms	—	REJECTED; approved amount ₹0; confidence 0.98.

TC008

Per-Claim Limit Exceeded ✓

Expected: REJECTED, reasons ['PER_CLAIM_EXCEEDED']

Got: REJECTED

Approved: ₹0 · **Confidence:** 0.98

Reasons given:

- Claimed amount ₹7,500 (eligible ₹7,500) exceeds the per-claim limit of ₹5,000 for CONSULTATION claims. Claims above this limit are not payable.

Line items:

ITEM	CLAIMED	VERDICT	REASON
Consultation Fee	₹2,000	✓ approved	—
Medicines	₹5,500	✓ approved	—

system_must checks:

- ✓ states limit and claimed amount

FULL TRACE — 19 STEPS

#	COMPONENT	ACTION	OUTCOME	TOOK	Δ CONF	DETAIL
1	intake_validator	validate submission payload	PASS	<1ms	—	Claim by EMP003 for CONSULTATION, treatment 2024-10-20, amount ₹7500, 2 document(s).
2	document_verifier	required document types	PASS	<1ms	—	All required types for CONSULTATION present (PRESCRIPTION, HOSPITAL_BILL). Uploaded: PRESCRIPTION, HOSPITAL_BILL.
3	extraction_agent	extract structured data from documents	SKIPPED	<1ms	—	Pre-extracted content supplied with the submission; vision extraction skipped (test mode).
4	document_verifier	patient consistency	SKIPPED	<1ms	—	No patient name could be read from any document; nothing to cross-check.
5	policy_retriever	rule source	PASS	<1ms	—	In-memory policy snapshot (authoritative; no policy graph configured).
6	adjudication_engine	eligibility.member	PASS	3ms	—	Member EMP003 (Amit Verma, SELF) found in roster.
7	adjudication_engine	eligibility.policy	PASS	<1ms	—	Policy PLUM_GHI_2024 active on treatment date 2024-10-20.
8	adjudication_engine	submission.deadline	PASS	<1ms	—	Submitted 0 days after treatment, within the 30-day deadline.
9	adjudication_engine	submission.minimum_amount	PASS	<1ms	—	Claimed amount ₹7,500 meets the ₹500 minimum.
10	adjudication_engine	amount_reconciliation	PASS	<1ms	—	Claimed ₹7,500 matches the documented ₹7,500 (itemized line items).
11	adjudication_engine	exclusions.claim_level	PASS	<1ms	—	Diagnosis/treatment ('Gastroenteritis') matches no policy exclusion.
12	adjudication_engine	waiting_period.initial	PASS	<1ms	—	Initial 30-day waiting period complete (joined 2024-04-01).
13	adjudication_engine	waiting_period.condition_specific	PASS	<1ms	—	Diagnosis 'Gastroenteritis' matches no condition-specific waiting period.
14	adjudication_engine	pre_authorization	PASS	<1ms	—	No pre-authorization rules apply to CONSULTATION claims.
15	adjudication_engine	line_items	PASS	<1ms	—	All 2 line item(s) covered under consultation.
16	adjudication_engine	per_claim_limit	FAIL	<1ms	—	Claimed amount ₹7,500 (eligible ₹7,500) exceeds the per-claim limit of ₹5,000 for CONSULTATION claims. Claims above this limit are not payable.

17	fraud_checker	velocity and threshold rules	PASS	<1ms	—	No fraud signals: same-day, monthly, and high-value checks all within limits.
18	decision_synthesizer	confidence rollup	PASS	<1ms	—	Confidence 0.98 = base 0.98; no penalties.
19	decision_synthesizer	final decision	FAIL	<1ms	—	REJECTED; approved amount ₹0; confidence 0.98.

TC009

Fraud Signal — Multiple Same-Day Claims ✓

Expected: MANUAL_REVIEW

Got: MANUAL_REVIEW

Approved: ₹0 · Confidence: 0.93

Reasons given:

- Routed to manual review — fraud signals require a human decision; the claim is not auto-rejected.
- This is claim #4 from member EMPO08 on 2024-10-30 (earlier today: CLM_0081, CLM_0082, CLM_0083) — above the same-day limit of 2.
- bill not itemized; adjudicated on billed total
- Rule checks alone would have approved ₹4,320; held pending review.
- Bill has no itemized line items; adjudicated against the billed total.
- YTD claims amount not provided; assumed ₹0 for the annual OPD limit check.

Line items:

ITEM	CLAIMED	VERDICT	REASON
Billed amount (no itemized bill)	₹4,800	✓ approved	—

Fraud signals:

- SAME_DAY_LIMIT_EXCEEDED — This is claim #4 from member EMPO08 on 2024-10-30 (earlier today: CLM_0081, CLM_0082, CLM_0083) — above the same-day limit of 2.

system_must checks:

- ✓ includes the specific fraud signals

FULL TRACE — 20 STEPS

#	COMPONENT	ACTION	OUTCOME	TOOK	Δ CONF	DETAIL
1	intake_validator	validate submission payload	PASS	<1ms	—	Claim by EMPO08 for CONSULTATION, treatment 2024-10-30, amount ₹4800, 2 document(s).
2	document_verifier	required document types	PASS	<1ms	—	All required types for CONSULTATION present (PRESCRIPTION, HOSPITAL_BILL). Uploaded: PRESCRIPTION, HOSPITAL_BILL.
3	extraction_agent	extract structured data from documents	SKIPPED	<1ms	—	Pre-extracted content supplied with the submission; vision extraction skipped (test mode).
4	document_verifier	patient consistency	SKIPPED	<1ms	—	No patient name could be read from any document; nothing to cross-check.
5	policy_retriever	rule source	PASS	<1ms	—	In-memory policy snapshot (authoritative; no policy graph configured).

6	adjudication_engine	eligibility.member	PASS	1ms	—	Member EMP008 (Ravi Menon, SELF) found in roster.
7	adjudication_engine	eligibility.policy	PASS	<1ms	—	Policy PLUM_GHI_2024 active on treatment date 2024-10-30.
8	adjudication_engine	submission.deadline	PASS	<1ms	—	Submitted 0 days after treatment, within the 30-day deadline.
9	adjudication_engine	submission.minimum_amount	PASS	<1ms	—	Claimed amount ₹4,800 meets the ₹500 minimum.
10	adjudication_engine	amount_reconciliation	PASS	<1ms	—	Claimed ₹4,800 matches the documented ₹4,800 (billed total).
11	adjudication_engine	exclusions.claim_level	PASS	<1ms	—	Diagnosis/treatment ('Migraine') matches no policy exclusion.
12	adjudication_engine	waiting_period.initial	PASS	<1ms	—	Initial 30-day waiting period complete (joined 2024-04-01).
13	adjudication_engine	waiting_period.condition_specific	PASS	<1ms	—	Diagnosis 'Migraine' matches no condition-specific waiting period.
14	adjudication_engine	pre_authorization	PASS	<1ms	—	No pre-authorization rules apply to CONSULTATION claims.
15	adjudication_engine	line_items	PASS	<1ms	—	All 1 line item(s) covered under consultation.
16	adjudication_engine	per_claim_limit	PASS	<1ms	—	Eligible amount ₹4,800 is within the ₹5,000 per-claim limit.
17	adjudication_engine	financial computation	PASS	<1ms	—	eligible_base ₹4800.00 → network_discount ₹4800.00 → copay ₹4320.00 → sub_limit_cap ₹4320.00 → annual_limit_cap ₹4320.00
18	fraud_checker	velocity and threshold rules	FAIL	<1ms	—	This is claim #4 from member EMP008 on 2024-10-30 (earlier today: CLM_0081, CLM_0082, CLM_0083) — above the same-day limit of 2.
19	decision_synthesizer	confidence rollup	PASS	<1ms	—	Confidence 0.93 = base 0.98 -0.05 (bill not itemized; adjudicated on billed total)
20	decision_synthesizer	final decision	PASS	<1ms	—	MANUAL_REVIEW; approved amount ₹0; confidence 0.93.

TC010

Network Hospital — Discount Applied ✓

Expected: APPROVED, ₹3,240

Got: APPROVED

Approved: ₹3,240 · **Confidence:** 0.98

Reasons given:

- Approved ₹3,240 of claimed ₹4,500.
- Network hospital discount (20%) applied. -₹900 (₹4,500 → ₹3,600)
- Co-pay (10%) borne by member. -₹360 (₹3,600 → ₹3,240)

Line items:

ITEM	CLAIMED	VERDICT	REASON
Consultation Fee	₹1,500	✓ approved	—
Medicines	₹3,000	✓ approved	—

system_must checks:

- ✓ shows discount and co-pay breakdown

FULL TRACE — 20 STEPS

#	COMPONENT	ACTION	OUTCOME	TOOK	Δ CONF	DETAIL
1	intake_validator	validate submission payload	PASS	<1ms	—	Claim by EMP010 for CONSULTATION, treatment 2024-11-03, amount ₹4500, 2 document(s).
2	document_verifier	required document types	PASS	<1ms	—	All required types for CONSULTATION present (PRESCRIPTION, HOSPITAL_BILL). Uploaded: PRESCRIPTION, HOSPITAL_BILL.
3	extraction_agent	extract structured data from documents	SKIPPED	<1ms	—	Pre-extracted content supplied with the submission; vision extraction skipped (test mode).
4	document_verifier	patient consistency	PASS	<1ms	—	All named documents agree on patient 'Deepak Shah'.
5	policy_retriever	rule source	PASS	<1ms	—	In-memory policy snapshot (authoritative; no policy graph configured).
6	adjudication_engine	eligibility.member	PASS	4ms	—	Member EMP010 (Deepak Shah, SELF) found in roster.
7	adjudication_engine	eligibility.policy	PASS	<1ms	—	Policy PLUM_GHI_2024 active on treatment date 2024-11-03.
8	adjudication_engine	submission.deadline	PASS	<1ms	—	Submitted 0 days after treatment, within the 30-day deadline.
9	adjudication_engine	submission.minimum_amount	PASS	<1ms	—	Claimed amount ₹4,500 meets the ₹500 minimum.
10	adjudication_engine	amount_reconciliation	PASS	<1ms	—	Claimed ₹4,500 matches the documented ₹4,500 (itemized line items).
11	adjudication_engine	exclusions.claim_level	PASS	<1ms	—	Diagnosis/treatment ('Acute Bronchitis') matches no policy exclusion.
12	adjudication_engine	waiting_period.initial	PASS	<1ms	—	Initial 30-day waiting period complete (joined 2024-04-01).
13	adjudication_engine	waiting_period.condition_specific	PASS	<1ms	—	Diagnosis 'Acute Bronchitis' matches no condition-specific waiting period.
14	adjudication_engine	pre_authorization	PASS	<1ms	—	No pre-authorization rules apply to CONSULTATION claims.
15	adjudication_engine	line_items	PASS	<1ms	—	All 2 line item(s) covered under consultation.

16	adjudication_engine	per_claim_limit	PASS	<1ms	—	Eligible amount ₹4,500 is within the ₹5,000 per-claim limit.
17	adjudication_engine	financial computation	PASS	<1ms	—	eligible_base ₹4500.00 → network_discount ₹3600.00 → copay ₹3240.00 → sub_limit_cap ₹3240.00 → annual_limit_cap ₹3240.00
18	fraud_checker	velocity and threshold rules	PASS	<1ms	—	No fraud signals: same-day, monthly, and high-value checks all within limits.
19	decision_synthesizer	confidence rollup	PASS	<1ms	—	Confidence 0.98 = base 0.98; no penalties.
20	decision_synthesizer	final decision	PASS	<1ms	—	APPROVED; approved amount ₹3,240; confidence 0.98.

TC011

Component Failure — Graceful Degradation ✓

Expected: APPROVED

Got: APPROVED

Approved: ₹4,000 · **Confidence:** 0.78 · **Manual review recommended**

Reasons given:

- Approved ₹4,000 of claimed ₹4,000.
- Processing was incomplete — fraud_checker failed and was skipped. Manual review recommended.
- fraud_checker failed and was skipped
- YTD claims amount not provided; assumed ₹0 for the annual OPD limit check.

Line items:

ITEM	CLAIMED	VERDICT	REASON
Panchakarma Therapy (5 sessions)	₹3,000	✓ approved	—
Consultation	₹1,000	✓ approved	—

Degraded components: fraud_checker

system_must checks:

- ✓ failure visible + review recommended

FULL TRACE — 20 STEPS

#	COMPONENT	ACTION	OUTCOME	TOOK	Δ CONF	DETAIL
1	intake_validator	validate submission payload	PASS	<1ms	—	Claim by EMPO06 for ALTERNATIVE_MEDICINE, treatment 2024-10-28, amount ₹4000, 2 document(s).
2	document_verifier	required document types	PASS	<1ms	—	All required types for ALTERNATIVE_MEDICINE present (PRESCRIPTION, HOSPITAL_BILL). Uploaded: PRESCRIPTION, HOSPITAL_BILL.
3	extraction_agent	extract structured data from documents	SKIPPED	1ms	—	Pre-extracted content supplied with the submission; vision extraction skipped (test mode).
4	document_verifier	patient consistency	SKIPPED	<1ms	—	No patient name could be read from any document; nothing to cross-check.

5	policy_retriever	rule source	PASS	<1ms	—	In-memory policy snapshot (authoritative; no policy graph configured).
6	adjudication_engine	eligibility.member	PASS	1ms	—	Member EMPO06 (Kavita Nair, SELF) found in roster.
7	adjudication_engine	eligibility.policy	PASS	<1ms	—	Policy PLUM_GHI_2024 active on treatment date 2024-10-28.
8	adjudication_engine	submission.deadline	PASS	<1ms	—	Submitted 0 days after treatment, within the 30-day deadline.
9	adjudication_engine	submission.minimum_amount	PASS	<1ms	—	Claimed amount ₹4,000 meets the ₹500 minimum.
10	adjudication_engine	amount_reconciliation	PASS	<1ms	—	Claimed ₹4,000 matches the documented ₹4,000 (itemized line items).
11	adjudication_engine	exclusions.claim_level	PASS	<1ms	—	Diagnosis/treatment ('Chronic Joint Pain') matches no policy exclusion.
12	adjudication_engine	waiting_period.initial	PASS	<1ms	—	Initial 30-day waiting period complete (joined 2024-04-01).
13	adjudication_engine	waiting_period.condition_specific	PASS	<1ms	—	Diagnosis 'Chronic Joint Pain' matches no condition-specific waiting period.
14	adjudication_engine	pre_authorization	PASS	<1ms	—	No pre-authorization rules apply to ALTERNATIVE_MEDICINE claims.
15	adjudication_engine	line_items	PASS	<1ms	—	All 2 line item(s) covered under alternative_medicine.
16	adjudication_engine	per_claim_limit	PASS	<1ms	—	Eligible amount ₹4,000 is within the ₹8,000 per-claim limit.
17	adjudication_engine	financial computation	PASS	<1ms	—	eligible_base ₹4000.00 → network_discount ₹4000.00 → copay ₹4000.00 → sub_limit_cap ₹4000.00 → annual_limit_cap ₹4000.00
18	fraud_checker	component failure handled	DEGRADED	<1ms	-0.20	Component failure (simulated): fraud checker crashed and was skipped; velocity rules were not evaluated for this claim.
19	decision_synthesizer	confidence rollup	PASS	<1ms	—	Confidence 0.78 = base 0.98 -0.20 (fraud_checker failed and was skipped)
20	decision_synthesizer	final decision	PASS	<1ms	—	APPROVED; approved amount ₹4,000; confidence 0.78.

TC012

Excluded Treatment ✓

Expected: REJECTED, reasons ['EXCLUDED_CONDITION']

Got: REJECTED

Approved: ₹0 · **Confidence:** 0.98

Reasons given:

- 'Morbid Obesity — BMI 37' matches policy exclusion 'Obesity and weight loss programs' (matched on: obesity). This condition is not covered under the policy at any time.

- Also failed: Diagnosis 'Morbid Obesity — BMI 37' falls under the 365-day waiting period for obesity treatment (member joined 2024-04-01). Treatment on 2024-10-18 is inside this period. The member is eligible for obesity treatment-related claims from 2025-04-01.

Line items:

ITEM	CLAIMED	VERDICT	REASON
Bariatric Consultation	₹3,000	✖ rejected	Claim excluded under policy: 'Obesity and weight loss programs'.
Personalised Diet and Nutrition Program	₹5,000	✖ rejected	Claim excluded under policy: 'Obesity and weight loss programs'.

FULL TRACE — 19 STEPS

#	COMPONENT	ACTION	OUTCOME	TOOK	Δ CONF	DETAIL
1	intake_validator	validate submission payload	PASS	<1ms	—	Claim by EMP009 for CONSULTATION, treatment 2024-10-18, amount ₹8000, 2 document(s).
2	document_verifier	required document types	PASS	<1ms	—	All required types for CONSULTATION present (PRESCRIPTION, HOSPITAL_BILL). Uploaded: PRESCRIPTION, HOSPITAL_BILL.
3	extraction_agent	extract structured data from documents	SKIPPED	<1ms	—	Pre-extracted content supplied with the submission; vision extraction skipped (test mode).
4	document_verifier	patient consistency	SKIPPED	<1ms	—	No patient name could be read from any document; nothing to cross-check.
5	policy_retriever	rule source	PASS	<1ms	—	In-memory policy snapshot (authoritative; no policy graph configured).
6	adjudication_engine	eligibility.member	PASS	<1ms	—	Member EMP009 (Anita Desai, SELF) found in roster.
7	adjudication_engine	eligibility.policy	PASS	<1ms	—	Policy PLUM_GHI_2024 active on treatment date 2024-10-18.
8	adjudication_engine	submission.deadline	PASS	<1ms	—	Submitted 0 days after treatment, within the 30-day deadline.
9	adjudication_engine	submission.minimum_amount	PASS	<1ms	—	Claimed amount ₹8,000 meets the ₹500 minimum.
10	adjudication_engine	amount_reconciliation	PASS	<1ms	—	Claimed ₹8,000 matches the documented ₹8,000 (itemized line items).
11	adjudication_engine	exclusions.claim_level	FAIL	<1ms	—	'Morbid Obesity — BMI 37' matches policy exclusion 'Obesity and weight loss programs' (matched on: obesity). This condition is not covered under the policy at any time.
12	adjudication_engine	waiting_period.initial	PASS	<1ms	—	Initial 30-day waiting period complete (joined 2024-04-01).

13	adjudication_engine	waiting_period.obesity_treatment	FAIL	<1ms	—	Diagnosis 'Morbid Obesity — BMI 37' falls under the 365-day waiting period for obesity treatment (member joined 2024-04-01). Treatment on 2024-10-18 is inside this period. The member is eligible for obesity treatment-related claims from 2025-04-01.
14	adjudication_engine	pre_authorization	PASS	<1ms	—	No pre-authorization rules apply to CONSULTATION claims.
15	adjudication_engine	line_items	FAIL	<1ms	—	All 2 line item(s) rejected — the claim is excluded.
16	adjudication_engine	per_claim_limit	PASS	<1ms	—	Eligible amount ₹0 is within the ₹5,000 per-claim limit.
17	fraud_checker	velocity and threshold rules	PASS	<1ms	—	No fraud signals: same-day, monthly, and high-value checks all within limits.
18	decision_synthesizer	confidence rollup	PASS	<1ms	—	Confidence 0.98 = base 0.98; no penalties.
19	decision_synthesizer	final decision	FAIL	<1ms	—	REJECTED; approved amount ₹0; confidence 0.98.

PART TWO

Upload path — the same 12 cases as real files

Documents are the generated fixtures in `data/mock_documents/`, uploaded with their type left on *auto-detect* so the classifier's own reading is what the requirement check sees. Confidence differs from the structured run by design: vision-read fields carry per-field scores, pre-supplied content does not.

TC001 · UPLOAD

Wrong Document Uploaded ✓

Expected: early stop, no decision
Got: — (stopped: document problems)

Member-facing problems:

- **[WRONG_TYPE]** A CONSULTATION claim requires: PRESCRIPTION and HOSPITAL_BILL. You uploaded PRESCRIPTION (prescription_rajesh.jpg), PRESCRIPTION (prescription_followup.jpg) — a PRESCRIPTION was provided where a HOSPITAL_BILL is required, and no HOSPITAL_BILL was included.
 - *Action:* Upload the hospital bill for this treatment and resubmit the claim.

system_must checks:

- ✓ names uploaded and required types

FULL TRACE — 5 STEPS

#	COMPONENT	ACTION	OUTCOME	TOOK	Δ CONF	DETAIL
1	intake_validator	validate submission payload	PASS	<1ms	—	Claim by EMPO01 for CONSULTATION, treatment 2024-11-01, amount ₹1500, 2 document(s).
2	document_verifier	document classification	PASS	8.8s	—	FO01: classified as PRESCRIPTION (conf 0.99), quality GOOD.
3	document_verifier	document classification	PASS	1.8s	—	FO02: classified as PRESCRIPTION (conf 0.99), quality GOOD.
4	document_verifier	required document types	FAIL	<1ms	—	Missing required type(s): HOSPITAL_BILL. Uploaded: PRESCRIPTION (prescription_rajesh.jpg), PRESCRIPTION (prescription_followup.jpg).
5	document_verifier	verification stopped	FAIL	<1ms	—	1 document problem(s) found; processing stopped before any decision. The claim is returned to the member with instructions.

TC002 · UPLOAD

Unreadable Document ✓

Expected: early stop, no decision
Got: — (stopped: document problems)

Member-facing problems:

- **[UNREADABLE]** 'pharmacy_bill_unreadable.jpg' is too blurred to read — no text could be recovered from it, so we cannot tell what document it is or what it says. Your claim has not been rejected and nothing else is wrong with it. Your other document is fine — only this one needs replacing.
 - *Action:* Re-upload 'pharmacy_bill_unreadable.jpg': photograph the original in good light with the whole page flat in frame, hold steady until the camera focuses, and check the text is readable before submitting. A scanned PDF works too.

system_must checks:

- ✓ asks re-upload of the specific document

FULL TRACE — 6 STEPS

#	COMPONENT	ACTION	OUTCOME	TOOK	Δ CONF	DETAIL
1	intake_validator	validate submission payload	PASS	<1ms	—	Claim by EMPO04 for PHARMACY, treatment 2024-10-25, amount ₹800, 2 document(s).
2	document_verifier	document classification	PASS	1.8s	—	FO01: classified as PRESCRIPTION (conf 0.99), quality GOOD.
3	document_verifier	image legibility	FAIL	51ms	—	FO02 ('pharmacy_bill_unreadable.jpg') measured focus score 0.42, below the legibility floor of 2.0 — the image carries no readable detail. Classification and extraction skipped; nothing can be read from it.
4	document_verifier	required document types	SKIPPED	<1ms	—	1 document(s) could not be read, so their type is unknown; the PHARMACY requirement (PRESCRIPTION, PHARMACY_BILL) is re-checked once a legible copy is uploaded.
5	document_verifier	readability check	FAIL	<1ms	—	'pharmacy_bill_unreadable.jpg' is unreadable. Claim not rejected; a re-upload of this one document is all that is required.
6	document_verifier	verification stopped	FAIL	<1ms	—	1 document problem(s) found; processing stopped before any decision. The claim is returned to the member with instructions.

TC003 · UPLOAD

Documents Belong to Different Patients ✓

Expected: early stop, no decision
Got: — (stopped: document problems)

Member-facing problems:

- **[PATIENT_MISMATCH]** The uploaded documents belong to different patients: PRESCRIPTION ('prescription_rajesh.jpg') is for Rajesh Kumar; HOSPITAL_BILL ('hospital_bill_arjun_mehta.jpg') is for Arjun Mehta. All documents in one claim must be for the same person.
 - *Action:* Check the files and re-upload documents that all belong to the patient this claim is for. If you are claiming for a dependent, every document must carry the dependent's name.

system_must checks:

- ✓ names both patients

FULL TRACE — 7 STEPS

#	COMPONENT	ACTION	OUTCOME	TOOK	Δ CONF	DETAIL
1	intake_validator	validate submission payload	PASS	<1ms	—	Claim by EMP001 for CONSULTATION, treatment 2024-11-01, amount ₹1500, 2 document(s).
2	document_verifier	document classification	PASS	2.0s	—	FO01: classified as PRESCRIPTION (conf 0.99), quality GOOD.
3	document_verifier	document classification	PASS	1.7s	—	FO02: classified as HOSPITAL_BILL (conf 0.99), quality GOOD.
4	document_verifier	required document types	PASS	<1ms	—	All required types for CONSULTATION present (PRESCRIPTION, HOSPITAL_BILL). Uploaded: PRESCRIPTION (prescription_rajesh.jpg), HOSPITAL_BILL (hospital_bill_arjun_mehta.jpg).
5	extraction_agent	extract structured data from documents	PASS	3.8s	—	2 document(s) extracted via GPT-4o vision; 0 used pre-extracted content (test mode).
6	document_verifier	patient consistency	FAIL	<1ms	—	Documents name different patients: PRESCRIPTION ('prescription_rajesh.jpg') is for Rajesh Kumar; HOSPITAL_BILL ('hospital_bill_arjun_mehta.jpg') is for Arjun Mehta.
7	document_verifier	verification stopped	FAIL	<1ms	—	1 document problem(s) found after reading the documents; processing stopped before any decision.

TC004 · UPLOAD

Clean Consultation — Full Approval ✓

Expected: APPROVED, ₹1,350

Got: APPROVED

Approved: ₹1,350 · Confidence: 0.98

Reasons given:

- Approved ₹1,350 of claimed ₹1,500.
- Co-pay (10%) borne by member. -₹150 (₹1,500 → ₹1,350)

Line items:

ITEM	CLAIMED	VERDICT	REASON
Consultation Fee (OPD)	₹1,000	✓ approved	—
CBC (Complete Blood Count)	₹300	✓ approved	—
Dengue NS1 Antigen Test	₹200	✓ approved	—

FULL TRACE — 22 STEPS

#	COMPONENT	ACTION	OUTCOME	TOOK	Δ CONF	DETAIL
1	intake_validator	validate submission payload	PASS	<1ms	—	Claim by EMP001 for CONSULTATION, treatment 2024-11-01, amount ₹1500, 2 document(s).

2	document_verifier	document classification	PASS	1.8s	—	FOO1: classified as PRESCRIPTION (conf 0.98), quality GOOD.
3	document_verifier	document classification	PASS	2.6s	—	FOO2: classified as HOSPITAL_BILL (conf 0.98), quality GOOD.
4	document_verifier	required document types	PASS	<1ms	—	All required types for CONSULTATION present (PRESCRIPTION, HOSPITAL_BILL). Uploaded: PRESCRIPTION (prescription_rajesh.jpg), HOSPITAL_BILL (hospital_bill_city_clinic.jpg).
5	extraction_agent	extract structured data from documents	PASS	2.9s	—	2 document(s) extracted via GPT-4o vision; 0 used pre-extracted content (test mode).
6	document_verifier	patient consistency	PASS	<1ms	—	All named documents agree on patient 'Rajesh Kumar'.
7	policy_retriever	rule source	PASS	<1ms	—	In-memory policy snapshot (authoritative; no policy graph configured).
8	adjudication_engine	eligibility.member	PASS	1ms	—	Member EMP001 (Rajesh Kumar, SELF) found in roster.
9	adjudication_engine	eligibility.policy	PASS	<1ms	—	Policy PLUM_GHI_2024 active on treatment date 2024-11-01.
10	adjudication_engine	submission.deadline	PASS	<1ms	—	Submitted 0 days after treatment, within the 30-day deadline.
11	adjudication_engine	submission.minimum_amount	PASS	<1ms	—	Claimed amount ₹1,500 meets the ₹500 minimum.
12	adjudication_engine	amount_reconciliation	PASS	<1ms	—	Claimed ₹1,500 matches the documented ₹1,500 (itemized line items).
13	adjudication_engine	exclusions.claim_level	PASS	<1ms	—	Diagnosis/treatment ('Viral Fever') matches no policy exclusion.
14	adjudication_engine	waiting_period.initial	PASS	<1ms	—	Initial 30-day waiting period complete (joined 2024-04-01).
15	adjudication_engine	waiting_period.condition_specific	PASS	<1ms	—	Diagnosis 'Viral Fever' matches no condition-specific waiting period.
16	adjudication_engine	pre_authorization	PASS	<1ms	—	No pre-authorization rules apply to CONSULTATION claims.
17	adjudication_engine	line_items	PASS	<1ms	—	All 3 line item(s) covered under consultation.
18	adjudication_engine	per_claim_limit	PASS	<1ms	—	Eligible amount ₹1,500 is within the ₹5,000 per-claim limit.
19	adjudication_engine	financial computation	PASS	<1ms	—	eligible_base ₹1500.00 → network_discount ₹1500.00 → copay ₹1350.00 → sub_limit_cap ₹1350.00 → annual_limit_cap ₹1350.00
20	fraud_checker	velocity and threshold rules	PASS	<1ms	—	No fraud signals: same-day, monthly, and high-value checks all within limits.
21	decision_synthesizer	confidence rollup	PASS	<1ms	—	Confidence 0.98 = base 0.98; no penalties.
22	decision_synthesizer	final decision	PASS	<1ms	—	APPROVED; approved amount ₹1,350; confidence 0.98.

TC005 · UPLOAD

Waiting Period — Diabetes ✓

Expected: REJECTED, reasons ['WAITING_PERIOD']

Got: REJECTED

Approved: ₹0 · **Confidence:** 0.98

Reasons given:

- Diagnosis 'Type 2 Diabetes Mellitus' falls under the 90-day waiting period for diabetes (member joined 2024-09-01). Treatment on 2024-10-15 is inside this period. The member is eligible for diabetes-related claims from 2024-11-30.

Line items:

ITEM	CLAIMED	VERDICT	REASON
Specialist Consultation	₹1,200	✓ approved	—
HbA1c Test	₹800	✓ approved	—
Fasting Blood Sugar	₹400	✓ approved	—
Lipid Profile	₹600	✓ approved	—

system_must checks:

- ✓ states the eligibility date

FULL TRACE — 21 STEPS

#	COMPONENT	ACTION	OUTCOME	TOOK	Δ CONF	DETAIL
1	intake_validator	validate submission payload	PASS	<1ms	—	Claim by EMP005 for CONSULTATION, treatment 2024-10-15, amount ₹3000, 2 document(s).
2	document_verifier	document classification	PASS	2.3s	—	FO01: classified as PRESCRIPTION (conf 0.99), quality GOOD.
3	document_verifier	document classification	PASS	1.9s	—	FO02: classified as HOSPITAL_BILL (conf 0.99), quality GOOD.
4	document_verifier	required document types	PASS	<1ms	—	All required types for CONSULTATION present (PRESCRIPTION, HOSPITAL_BILL). Uploaded: PRESCRIPTION (prescription_vikram_diabetes.jpg), HOSPITAL_BILL (hospital_bill_vikram.jpg).
5	extraction_agent	extract structured data from documents	PASS	3.7s	—	2 document(s) extracted via GPT-4o vision; 0 used pre-extracted content (test mode).
6	document_verifier	patient consistency	PASS	<1ms	—	All named documents agree on patient 'Vikram Joshi'.
7	policy_retriever	rule source	PASS	<1ms	—	In-memory policy snapshot (authoritative; no policy graph configured).
8	adjudication_engine	eligibility.member	PASS	1ms	—	Member EMP005 (Vikram Joshi, SELF) found in roster.

9	adjudication_engine	eligibility.policy	PASS	<1ms	—	Policy PLUM_GHI_2024 active on treatment date 2024-10-15.
10	adjudication_engine	submission.deadline	PASS	<1ms	—	Submitted 0 days after treatment, within the 30-day deadline.
11	adjudication_engine	submission.minimum_amount	PASS	<1ms	—	Claimed amount ₹3,000 meets the ₹500 minimum.
12	adjudication_engine	amount_reconciliation	PASS	<1ms	—	Claimed ₹3,000 matches the documented ₹3,000 (itemized line items).
13	adjudication_engine	exclusions.claim_level	PASS	<1ms	—	Diagnosis/treatment ('Type 2 Diabetes Mellitus') matches no policy exclusion.
14	adjudication_engine	waiting_period.initial	PASS	<1ms	—	Initial 30-day waiting period complete (joined 2024-09-01).
15	adjudication_engine	waiting_period.diabetes	FAIL	<1ms	—	Diagnosis 'Type 2 Diabetes Mellitus' falls under the 90-day waiting period for diabetes (member joined 2024-09-01). Treatment on 2024-10-15 is inside this period. The member is eligible for diabetes-related claims from 2024-11-30.
16	adjudication_engine	pre_authorization	PASS	<1ms	—	No pre-authorization rules apply to CONSULTATION claims.
17	adjudication_engine	line_items	PASS	<1ms	—	All 4 line item(s) covered under consultation.
18	adjudication_engine	per_claim_limit	PASS	<1ms	—	Eligible amount ₹3,000 is within the ₹5,000 per-claim limit.
19	fraud_checker	velocity and threshold rules	PASS	<1ms	—	No fraud signals: same-day, monthly, and high-value checks all within limits.
20	decision_synthesizer	confidence rollup	PASS	<1ms	—	Confidence 0.98 = base 0.98; no penalties.
21	decision_synthesizer	final decision	FAIL	<1ms	—	REJECTED; approved amount ₹0; confidence 0.98.

TC006 · UPLOAD

Dental Partial Approval — Cosmetic Exclusion ✓

Expected: PARTIAL, ₹8,000

Got: PARTIAL

Approved: ₹8,000 · Confidence: 0.98

Reasons given:

- Partially approved ₹8,000 of claimed ₹12,000.
- Line item rejected — 'Teeth Whitening' is an excluded procedure under the dental category ('Teeth Whitening').
- YTD claims amount not provided; assumed ₹0 for the annual OPD limit check.

Line items:

ITEM	CLAIMED	VERDICT	REASON
Root Canal Treatment	₹8,000	✓ approved	—
Teeth Whitening	₹4,000	✗ rejected	'Teeth Whitening' is an excluded procedure under the dental category ('Teeth Whitening').

system_must checks:

- ✓ itemizes line-item verdicts

FULL TRACE — 20 STEPS

#	COMPONENT	ACTION	OUTCOME	TOOK	Δ CONF	DETAIL
1	intake_validator	validate submission payload	PASS	<1ms	—	Claim by EMPO02 for DENTAL, treatment 2024-10-15, amount ₹12000, 1 document(s).
2	document_verifier	document classification	PASS	1.9s	—	FO01: classified as HOSPITAL_BILL (conf 0.98), quality GOOD.
3	document_verifier	required document types	PASS	<1ms	—	All required types for DENTAL present (HOSPITAL_BILL). Uploaded: HOSPITAL_BILL (dental_bill_priya.jpg).
4	extraction_agent	extract structured data from documents	PASS	3.2s	—	1 document(s) extracted via GPT-4o vision; 0 used pre-extracted content (test mode).
5	document_verifier	patient consistency	PASS	<1ms	—	All named documents agree on patient 'Priya Singh'.
6	policy_retriever	rule source	PASS	<1ms	—	In-memory policy snapshot (authoritative; no policy graph configured).
7	adjudication_engine	eligibility.member	PASS	<1ms	—	Member EMPO02 (Priya Singh, SELF) found in roster.
8	adjudication_engine	eligibility.policy	PASS	<1ms	—	Policy PLUM_GHI_2024 active on treatment date 2024-10-15.
9	adjudication_engine	submission.deadline	PASS	<1ms	—	Submitted 0 days after treatment, within the 30-day deadline.
10	adjudication_engine	submission.minimum_amount	PASS	<1ms	—	Claimed amount ₹12,000 meets the ₹500 minimum.
11	adjudication_engine	amount_reconciliation	PASS	<1ms	—	Claimed ₹12,000 matches the documented ₹12,000 (itemized line items).
12	adjudication_engine	exclusions.claim_level	SKIPPED	<1ms	—	No diagnosis or treatment text available to screen against exclusions.
13	adjudication_engine	waiting_period.initial	PASS	<1ms	—	Initial 30-day waiting period complete (joined 2024-04-01).
14	adjudication_engine	pre_authorization	PASS	<1ms	—	No pre-authorization rules apply to DENTAL claims.
15	adjudication_engine	line_items	FAIL	<1ms	—	1 of 2 line item(s) approved; 'Teeth Whitening' rejected: 'Teeth Whitening' is an excluded procedure under the dental category ('Teeth Whitening').
16	adjudication_engine	per_claim_limit	PASS	<1ms	—	Eligible amount ₹8,000 is within the ₹10,000 per-claim limit.
17	adjudication_engine	financial computation	PASS	<1ms	—	eligible_base ₹8000.00 → network_discount ₹8000.00 → copay ₹8000.00 → sub_limit_cap ₹8000.00 → annual_limit_cap ₹8000.00

18	fraud_checker	velocity and threshold rules	PASS	<1ms	—	No fraud signals: same-day, monthly, and high-value checks all within limits.
19	decision_synthesizer	confidence rollup	PASS	<1ms	—	Confidence 0.98 = base 0.98; no penalties.
20	decision_synthesizer	final decision	PASS	<1ms	—	PARTIAL; approved amount ₹8,000; confidence 0.98.

TC007 · UPLOAD

MRI Without Pre-Authorization ✓

Expected: REJECTED, reasons ['PRE_AUTH_MISSING']

Got: REJECTED

Approved: ₹0 · **Confidence:** 0.93

Reasons given:

- 'MRI Lumbar Spine' (₹15,000) requires pre-authorization because MRI above ₹10,000 must be pre-approved, and no authorization was obtained. To resubmit: request pre-authorization from the insurer for this procedure and attach the approval reference (valid 30 days from issue).
- Confidence 0.93 (from 0.98); the following were noted.
- Registration 'AP67890/2017' on F001 does not match known state formats; unverified.
- Also failed: Claimed amount ₹15,000 (eligible ₹15,000) exceeds the per-claim limit of ₹10,000 for DIAGNOSTIC claims. Claims above this limit are not payable.

Line items:

ITEM	CLAIMED	VERDICT	REASON
MRI Lumbar Spine	₹15,000	✓ approved	—

system_must checks:

- ✓ explains pre-auth and resubmission

FULL TRACE — 23 STEPS

#	COMPONENT	ACTION	OUTCOME	TOOK	Δ CONF	DETAIL
1	intake_validator	validate submission payload	PASS	<1ms	—	Claim by EMP007 for DIAGNOSTIC, treatment 2024-11-02, amount ₹15000, 3 document(s).
2	document_verifier	document classification	PASS	1.6s	—	F001: classified as PRESCRIPTION (conf 0.95), quality GOOD.
3	document_verifier	document classification	PASS	1.5s	—	F002: classified as DIAGNOSTIC_REPORT (conf 0.95), quality GOOD.
4	document_verifier	document classification	PASS	1.8s	—	F003: classified as HOSPITAL_BILL (conf 0.98), quality GOOD.
5	document_verifier	required document types	PASS	<1ms	—	All required types for DIAGNOSTIC present (PRESCRIPTION, LAB_REPORT, HOSPITAL_BILL). Uploaded: PRESCRIPTION (prescription_suresh_mri.jpg), DIAGNOSTIC_REPORT (lab_report_suresh_mri.jpg), HOSPITAL_BILL (hospital_bill_suresh_mri.jpg).

6	extraction_agent	extract structured data from documents	PASS	3.1s	—	3 document(s) extracted via GPT-4o vision; 0 used pre-extracted content (test mode).
7	document_verifier	patient consistency	PASS	<1ms	—	All named documents agree on patient 'Suresh Patil'.
8	document_verifier	registration format	DEGRADED	<1ms	-0.05	Registration 'AP67890/2017' on F001 does not match known state formats; unverified.
9	policy_retriever	rule source	PASS	<1ms	—	In-memory policy snapshot (authoritative; no policy graph configured).
10	adjudication_engine	eligibility.member	PASS	<1ms	—	Member EMP007 (Suresh Patil, SELF) found in roster.
11	adjudication_engine	eligibility.policy	PASS	<1ms	—	Policy PLUM_GHI_2024 active on treatment date 2024-11-02.
12	adjudication_engine	submission.deadline	PASS	<1ms	—	Submitted 0 days after treatment, within the 30-day deadline.
13	adjudication_engine	submission.minimum_amount	PASS	<1ms	—	Claimed amount ₹15,000 meets the ₹500 minimum.
14	adjudication_engine	amount_reconciliation	PASS	<1ms	—	Claimed ₹15,000 matches the documented ₹15,000 (itemized line items).
15	adjudication_engine	exclusions.claim_level	PASS	<1ms	—	Diagnosis/treatment ('Suspected Lumbar Disc Herniation') matches no policy exclusion.
16	adjudication_engine	waiting_period.initial	PASS	<1ms	—	Initial 30-day waiting period complete (joined 2024-04-01).
17	adjudication_engine	waiting_period.condition_specific	PASS	<1ms	—	Diagnosis 'Suspected Lumbar Disc Herniation' matches no condition-specific waiting period.
18	adjudication_engine	pre_authorization	FAIL	<1ms	—	'MRI Lumbar Spine' (₹15,000) requires pre-authorization because MRI above ₹10,000 must be pre-approved, and no authorization was obtained. To resubmit: request pre-authorization from the insurer for this procedure and attach the approval reference (valid 30 days from issue).
19	adjudication_engine	line_items	PASS	<1ms	—	All 1 line item(s) covered under diagnostic.
20	adjudication_engine	per_claim_limit	FAIL	<1ms	—	Claimed amount ₹15,000 (eligible ₹15,000) exceeds the per-claim limit of ₹10,000 for DIAGNOSTIC claims. Claims above this limit are not payable.
21	fraud_checker	velocity and threshold rules	PASS	<1ms	—	No fraud signals: same-day, monthly, and high-value checks all within limits.
22	decision_synthesizer	confidence rollup	PASS	<1ms	—	Confidence 0.93 = base 0.98 -0.05 (Registration 'AP67890/2017' on F001 does not match known state formats; unverified.)

23

decision_synthesizer

final decision

FAIL

<1ms

—

REJECTED; approved amount ₹0; confidence 0.93.

TC008 · UPLOAD

Per-Claim Limit Exceeded ✓

Expected: REJECTED, reasons ['PER_CLAIM_EXCEEDED']

Got: REJECTED

Approved: ₹0 · **Confidence:** 0.98

Reasons given:

- Claimed amount ₹7,500 (eligible ₹7,500) exceeds the per-claim limit of ₹5,000 for CONSULTATION claims. Claims above this limit are not payable.

Line items:

ITEM	CLAIMED	VERDICT	REASON
Consultation Fee	₹2,000	✓ approved	—
Medicines	₹5,500	✓ approved	—

system_must checks:

- ✓ states limit and claimed amount

FULL TRACE — 21 STEPS

#	COMPONENT	ACTION	OUTCOME	TOOK	Δ CONF	DETAIL
1	intake_validator	validate submission payload	PASS	<1ms	—	Claim by EMPO03 for CONSULTATION, treatment 2024-10-20, amount ₹7500, 2 document(s).
2	document_verifier	document classification	PASS	2.9s	—	FO01: classified as PRESCRIPTION (conf 0.96), quality GOOD.
3	document_verifier	document classification	PASS	2.4s	—	FO02: classified as HOSPITAL_BILL (conf 0.99), quality GOOD.
4	document_verifier	required document types	PASS	<1ms	—	All required types for CONSULTATION present (PRESCRIPTION, HOSPITAL_BILL). Uploaded: PRESCRIPTION (prescription_amit_gastro.jpg), HOSPITAL_BILL (hospital_bill_amit.jpg).
5	extraction_agent	extract structured data from documents	PASS	3.7s	—	2 document(s) extracted via GPT-4o vision; 0 used pre-extracted content (test mode).
6	document_verifier	patient consistency	PASS	<1ms	—	All named documents agree on patient 'Amit Verma'.
7	policy_retriever	rule source	PASS	<1ms	—	In-memory policy snapshot (authoritative; no policy graph configured).
8	adjudication_engine	eligibility.member	PASS	<1ms	—	Member EMPO03 (Amit Verma, SELF) found in roster.
9	adjudication_engine	eligibility.policy	PASS	<1ms	—	Policy PLUM_GHI_2024 active on treatment date 2024-10-20.
10	adjudication_engine	submission.deadline	PASS	<1ms	—	Submitted 0 days after treatment, within the 30-day deadline.

11	adjudication_engine	submission.minimum_amount	PASS	<1ms	—	Claimed amount ₹7,500 meets the ₹500 minimum.
12	adjudication_engine	amount_reconciliation	PASS	<1ms	—	Claimed ₹7,500 matches the documented ₹7,500 (itemized line items).
13	adjudication_engine	exclusions.claim_level	PASS	<1ms	—	Diagnosis/treatment ('Acute Gastroenteritis') matches no policy exclusion.
14	adjudication_engine	waiting_period.initial	PASS	<1ms	—	Initial 30-day waiting period complete (joined 2024-04-01).
15	adjudication_engine	waiting_period.condition_specific	PASS	<1ms	—	Diagnosis 'Acute Gastroenteritis' matches no condition-specific waiting period.
16	adjudication_engine	pre_authorization	PASS	<1ms	—	No pre-authorization rules apply to CONSULTATION claims.
17	adjudication_engine	line_items	PASS	<1ms	—	All 2 line item(s) covered under consultation.
18	adjudication_engine	per_claim_limit	FAIL	<1ms	—	Claimed amount ₹7,500 (eligible ₹7,500) exceeds the per-claim limit of ₹5,000 for CONSULTATION claims. Claims above this limit are not payable.
19	fraud_checker	velocity and threshold rules	PASS	<1ms	—	No fraud signals: same-day, monthly, and high-value checks all within limits.
20	decision_synthesizer	confidence rollup	PASS	<1ms	—	Confidence 0.98 = base 0.98; no penalties.
21	decision_synthesizer	final decision	FAIL	<1ms	—	REJECTED; approved amount ₹0; confidence 0.98.

TC009 · UPLOAD

Fraud Signal — Multiple Same-Day Claims ✓

Expected: MANUAL_REVIEW

Got: MANUAL_REVIEW

Approved: ₹0 · Confidence: 0.98

Reasons given:

- Routed to manual review — fraud signals require a human decision; the claim is not auto-rejected.
- This is claim #4 from member EMPO08 on 2024-10-30 (earlier today: CLM_0081, CLM_0082, CLM_0083) — above the same-day limit of 2.
- Rule checks alone would have approved ₹4,320; held pending review.
- YTD claims amount not provided; assumed ₹0 for the annual OPD limit check.

Line items:

ITEM	CLAIMED	VERDICT	REASON
Specialist Consultation	₹1,800	✓ approved	—
Medicines	₹3,000	✓ approved	—

Fraud signals:

- SAME_DAY_LIMIT_EXCEEDED — This is claim #4 from member EMPO08 on 2024-10-30 (earlier today: CLM_0081, CLM_0082, CLM_0083) — above the same-day limit of 2.

system_must checks:

- ✓ includes the specific fraud signals

FULL TRACE — 22 STEPS

#	COMPONENT	ACTION	OUTCOME	TOOK	Δ CONF	DETAIL
1	intake_validator	validate submission payload	PASS	<1ms	—	Claim by EMP008 for CONSULTATION, treatment 2024-10-30, amount ₹4800, 2 document(s).
2	document_verifier	document classification	PASS	1.4s	—	FO01: classified as PRESCRIPTION (conf 0.98), quality GOOD.
3	document_verifier	document classification	PASS	2.7s	—	FO02: classified as HOSPITAL_BILL (conf 0.99), quality GOOD.
4	document_verifier	required document types	PASS	<1ms	—	All required types for CONSULTATION present (PRESCRIPTION, HOSPITAL_BILL). Uploaded: PRESCRIPTION (prescription_ravi_migraine.jpg), HOSPITAL_BILL (hospital_bill_ravi.jpg).
5	extraction_agent	extract structured data from documents	PASS	3.8s	—	2 document(s) extracted via GPT-4o vision; 0 used pre-extracted content (test mode).
6	document_verifier	patient consistency	PASS	<1ms	—	All named documents agree on patient 'Ravi Menon'.
7	policy_retriever	rule source	PASS	<1ms	—	In-memory policy snapshot (authoritative; no policy graph configured).
8	adjudication_engine	eligibility.member	PASS	1ms	—	Member EMP008 (Ravi Menon, SELF) found in roster.
9	adjudication_engine	eligibility.policy	PASS	<1ms	—	Policy PLUM_GHI_2024 active on treatment date 2024-10-30.
10	adjudication_engine	submission.deadline	PASS	<1ms	—	Submitted 0 days after treatment, within the 30-day deadline.
11	adjudication_engine	submission.minimum_amount	PASS	<1ms	—	Claimed amount ₹4,800 meets the ₹500 minimum.
12	adjudication_engine	amount_reconciliation	PASS	<1ms	—	Claimed ₹4,800 matches the documented ₹4,800 (itemized line items).
13	adjudication_engine	exclusions.claim_level	PASS	<1ms	—	Diagnosis/treatment ('Migraine without aura') matches no policy exclusion.
14	adjudication_engine	waiting_period.initial	PASS	<1ms	—	Initial 30-day waiting period complete (joined 2024-04-01).
15	adjudication_engine	waiting_period.condition_specific	PASS	<1ms	—	Diagnosis 'Migraine without aura' matches no condition-specific waiting period.
16	adjudication_engine	pre_authorization	PASS	<1ms	—	No pre-authorization rules apply to CONSULTATION claims.

17	adjudication_engine	line_items	PASS	<1ms	—	All 2 line item(s) covered under consultation.
18	adjudication_engine	per_claim_limit	PASS	<1ms	—	Eligible amount ₹4,800 is within the ₹5,000 per-claim limit.
19	adjudication_engine	financial computation	PASS	<1ms	—	eligible_base ₹4800.00 → network_discount ₹4800.00 → copay ₹4320.00 → sub_limit_cap ₹4320.00 → annual_limit_cap ₹4320.00
20	fraud_checker	velocity and threshold rules	FAIL	<1ms	—	This is claim #4 from member EMP008 on 2024-10-30 (earlier today: CLM_0081, CLM_0082, CLM_0083) — above the same-day limit of 2.
21	decision_synthesizer	confidence rollup	PASS	<1ms	—	Confidence 0.98 = base 0.98; no penalties.
22	decision_synthesizer	final decision	PASS	<1ms	—	MANUAL_REVIEW; approved amount ₹0; confidence 0.98.

TC010 · UPLOAD

Network Hospital — Discount Applied ✓

Expected: APPROVED, ₹3,240

Got: APPROVED

Approved: ₹3,240 · **Confidence:** 0.98

Reasons given:

- Approved ₹3,240 of claimed ₹4,500.
- Network hospital discount (20%) applied. -₹900 (₹4,500 → ₹3,600)
- Co-pay (10%) borne by member. -₹360 (₹3,600 → ₹3,240)

Line items:

ITEM	CLAIMED	VERDICT	REASON
Consultation Fee	₹1,500	✓ approved	—
Medicines	₹3,000	✓ approved	—

system_must checks:

- ✓ shows discount and co-pay breakdown

FULL TRACE — 22 STEPS

#	COMPONENT	ACTION	OUTCOME	TOOK	Δ CONF	DETAIL
1	intake_validator	validate submission payload	PASS	<1ms	—	Claim by EMP010 for CONSULTATION, treatment 2024-11-03, amount ₹4500, 2 document(s).
2	document_verifier	document classification	PASS	2.8s	—	FO01: classified as PRESCRIPTION (conf 0.99), quality GOOD.
3	document_verifier	document classification	PASS	3.6s	—	FO02: classified as HOSPITAL_BILL (conf 0.98), quality GOOD.
4	document_verifier	required document types	PASS	<1ms	—	All required types for CONSULTATION present (PRESCRIPTION, HOSPITAL_BILL). Uploaded: PRESCRIPTION (prescription_deepak_apollo.jpg), HOSPITAL_BILL (hospital_bill_apollo_deepak.jpg).

5	extraction_agent	extract structured data from documents	PASS	4.8s	—	2 document(s) extracted via GPT-4o vision; 0 used pre-extracted content (test mode).
6	document_verifier	patient consistency	PASS	<1ms	—	All named documents agree on patient 'Deepak Shah'.
7	policy_retriever	rule source	PASS	<1ms	—	In-memory policy snapshot (authoritative; no policy graph configured).
8	adjudication_engine	eligibility.member	PASS	1ms	—	Member EMP010 (Deepak Shah, SELF) found in roster.
9	adjudication_engine	eligibility.policy	PASS	<1ms	—	Policy PLUM_GHI_2024 active on treatment date 2024-11-03.
10	adjudication_engine	submission.deadline	PASS	<1ms	—	Submitted 0 days after treatment, within the 30-day deadline.
11	adjudication_engine	submission.minimum_amount	PASS	<1ms	—	Claimed amount ₹4,500 meets the ₹500 minimum.
12	adjudication_engine	amount_reconciliation	PASS	<1ms	—	Claimed ₹4,500 matches the documented ₹4,500 (itemized line items).
13	adjudication_engine	exclusions.claim_level	PASS	<1ms	—	Diagnosis/treatment ('Acute Bronchitis') matches no policy exclusion.
14	adjudication_engine	waiting_period.initial	PASS	<1ms	—	Initial 30-day waiting period complete (joined 2024-04-01).
15	adjudication_engine	waiting_period.condition_specific	PASS	<1ms	—	Diagnosis 'Acute Bronchitis' matches no condition-specific waiting period.
16	adjudication_engine	pre_authorization	PASS	<1ms	—	No pre-authorization rules apply to CONSULTATION claims.
17	adjudication_engine	line_items	PASS	<1ms	—	All 2 line item(s) covered under consultation.
18	adjudication_engine	per_claim_limit	PASS	<1ms	—	Eligible amount ₹4,500 is within the ₹5,000 per-claim limit.
19	adjudication_engine	financial computation	PASS	<1ms	—	eligible_base ₹4500.00 → network_discount ₹3600.00 → copay ₹3240.00 → sub_limit_cap ₹3240.00 → annual_limit_cap ₹3240.00
20	fraud_checker	velocity and threshold rules	PASS	<1ms	—	No fraud signals: same-day, monthly, and high-value checks all within limits.
21	decision_synthesizer	confidence rollup	PASS	<1ms	—	Confidence 0.98 = base 0.98; no penalties.
22	decision_synthesizer	final decision	PASS	<1ms	—	APPROVED; approved amount ₹3,240; confidence 0.98.

TC011 · UPLOAD

Component Failure — Graceful Degradation ✓

Expected: APPROVED

Got: APPROVED

Approved: ₹4,000 · Confidence: 0.78 · Manual review recommended

Reasons given:

- Approved ₹4,000 of claimed ₹4,000.
- Processing was incomplete — fraud_checker failed and was skipped. Manual review recommended.
- fraud_checker failed and was skipped

- YTD claims amount not provided; assumed ₹0 for the annual OPD limit check.

Line items:

ITEM	CLAIMED	VERDICT	REASON
Panchakarma Therapy	₹3,000	✓ approved	—
Consultation	₹1,000	✓ approved	—

Degraded components: fraud_checker

system_must checks:

- ✓ failure visible + review recommended

FULL TRACE — 22 STEPS

#	COMPONENT	ACTION	OUTCOME	TOOK	Δ CONF	DETAIL
1	intake_validator	validate submission payload	PASS	<1ms	—	Claim by EMPO06 for ALTERNATIVE_MEDICINE, treatment 2024-10-28, amount ₹4000, 2 document(s).
2	document_verifier	document classification	PASS	2.9s	—	FO01: classified as PRESCRIPTION (conf 0.95), quality GOOD.
3	document_verifier	document classification	PASS	2.7s	—	FO02: classified as HOSPITAL_BILL (conf 0.99), quality GOOD.
4	document_verifier	required document types	PASS	<1ms	—	All required types for ALTERNATIVE_MEDICINE present (PRESCRIPTION, HOSPITAL_BILL). Uploaded: PRESCRIPTION (prescription_kavita_ayurveda.jpg), HOSPITAL_BILL (hospital_bill_ayur_wellness.jpg).
5	extraction_agent	extract structured data from documents	PASS	3.5s	—	2 document(s) extracted via GPT-4o vision; 0 used pre-extracted content (test mode).
6	document_verifier	patient consistency	PASS	<1ms	—	All named documents agree on patient 'Kavita Nair'.
7	policy_retriever	rule source	PASS	<1ms	—	In-memory policy snapshot (authoritative; no policy graph configured).
8	adjudication_engine	eligibility.member	PASS	<1ms	—	Member EMPO06 (Kavita Nair, SELF) found in roster.
9	adjudication_engine	eligibility.policy	PASS	<1ms	—	Policy PLUM_GHI_2024 active on treatment date 2024-10-28.
10	adjudication_engine	submission.deadline	PASS	<1ms	—	Submitted 0 days after treatment, within the 30-day deadline.
11	adjudication_engine	submission.minimum_amount	PASS	<1ms	—	Claimed amount ₹4,000 meets the ₹500 minimum.
12	adjudication_engine	amount_reconciliation	PASS	<1ms	—	Claimed ₹4,000 matches the documented ₹4,000 (itemized line items).

13	adjudication_engine	exclusions.claim_level	PASS	<1ms	—	Diagnosis/treatment ('Chronic Joint Pain (Sandhigata Vata)') matches no policy exclusion.
14	adjudication_engine	waiting_period.initial	PASS	<1ms	—	Initial 30-day waiting period complete (joined 2024-04-01).
15	adjudication_engine	waiting_period.condition_specific	PASS	<1ms	—	Diagnosis 'Chronic Joint Pain (Sandhigata Vata)' matches no condition-specific waiting period.
16	adjudication_engine	pre_authorization	PASS	<1ms	—	No pre-authorization rules apply to ALTERNATIVE_MEDICINE claims.
17	adjudication_engine	line_items	PASS	<1ms	—	All 2 line item(s) covered under alternative_medicine.
18	adjudication_engine	per_claim_limit	PASS	<1ms	—	Eligible amount ₹4,000 is within the ₹8,000 per-claim limit.
19	adjudication_engine	financial computation	PASS	<1ms	—	eligible_base ₹4000.00 → network_discount ₹4000.00 → copay ₹4000.00 → sub_limit_cap ₹4000.00 → annual_limit_cap ₹4000.00
20	fraud_checker	component failure handled	DEGRADED	<1ms	-0.20	Component failure (simulated): fraud checker crashed and was skipped; velocity rules were not evaluated for this claim.
21	decision_synthesizer	confidence rollup	PASS	<1ms	—	Confidence 0.78 = base 0.98 -0.20 (fraud_checker failed and was skipped)
22	decision_synthesizer	final decision	PASS	<1ms	—	APPROVED; approved amount ₹4,000; confidence 0.78.

TC012 · UPLOAD

Excluded Treatment ✓

Expected: REJECTED, reasons ['EXCLUDED_CONDITION']

Got: REJECTED

Approved: ₹0 · **Confidence:** 0.98

Reasons given:

- 'Morbid Obesity - BMI 37' matches policy exclusion 'Obesity and weight loss programs' (matched on: obesity). This condition is not covered under the policy at any time.
- Also failed: Diagnosis 'Morbid Obesity - BMI 37' falls under the 365-day waiting period for obesity treatment (member joined 2024-04-01). Treatment on 2024-10-18 is inside this period. The member is eligible for obesity treatment-related claims from 2025-04-01.

Line items:

ITEM	CLAIMED	VERDICT	REASON
Bariatric Consultation	₹3,000	✗ rejected	Claim excluded under policy: 'Obesity and weight loss programs'.
Personalised Diet and Nutrition Program	₹5,000	✗ rejected	Claim excluded under policy: 'Obesity and weight loss programs'.

FULL TRACE — 21 STEPS

#	COMPONENT	ACTION	OUTCOME	TOOK	Δ CONF	DETAIL
1	intake_validator	validate submission payload	PASS	<1ms	—	Claim by EMP009 for CONSULTATION, treatment 2024-10-18, amount ₹8000, 2 document(s).
2	document_verifier	document classification	PASS	2.1s	—	FO01: classified as PRESCRIPTION (conf 0.95), quality GOOD.
3	document_verifier	document classification	PASS	1.6s	—	FO02: classified as HOSPITAL_BILL (conf 0.99), quality GOOD.
4	document_verifier	required document types	PASS	<1ms	—	All required types for CONSULTATION present (PRESCRIPTION, HOSPITAL_BILL). Uploaded: PRESCRIPTION (prescription_anita_bariatric.jpg), HOSPITAL_BILL (hospital_bill_anita_bariatric.jpg).
5	extraction_agent	extract structured data from documents	PASS	3.9s	—	2 document(s) extracted via GPT-4o vision; 0 used pre-extracted content (test mode).
6	document_verifier	patient consistency	PASS	<1ms	—	All named documents agree on patient 'Anita Desai'.
7	policy_retriever	rule source	PASS	<1ms	—	In-memory policy snapshot (authoritative; no policy graph configured).
8	adjudication_engine	eligibility.member	PASS	<1ms	—	Member EMP009 (Anita Desai, SELF) found in roster.
9	adjudication_engine	eligibility.policy	PASS	<1ms	—	Policy PLUM_GHI_2024 active on treatment date 2024-10-18.
10	adjudication_engine	submission.deadline	PASS	<1ms	—	Submitted 0 days after treatment, within the 30-day deadline.
11	adjudication_engine	submission.minimum_amount	PASS	<1ms	—	Claimed amount ₹8,000 meets the ₹500 minimum.
12	adjudication_engine	amount_reconciliation	PASS	<1ms	—	Claimed ₹8,000 matches the documented ₹8,000 (itemized line items).
13	adjudication_engine	exclusions.claim_level	FAIL	<1ms	—	'Morbid Obesity – BMI 37' matches policy exclusion 'Obesity and weight loss programs' (matched on: obesity). This condition is not covered under the policy at any time.
14	adjudication_engine	waiting_period.initial	PASS	<1ms	—	Initial 30-day waiting period complete (joined 2024-04-01).

15	adjudication_engine	waiting_period.obesity_treatment	FAIL	<1ms	—	Diagnosis 'Morbid Obesity - BMI 37' falls under the 365-day waiting period for obesity treatment (member joined 2024-04-01). Treatment on 2024-10-18 is inside this period. The member is eligible for obesity treatment-related claims from 2025-04-01.
16	adjudication_engine	pre_authorization	PASS	<1ms	—	No pre-authorization rules apply to CONSULTATION claims.
17	adjudication_engine	line_items	FAIL	<1ms	—	All 2 line item(s) rejected — the claim is excluded.
18	adjudication_engine	per_claim_limit	PASS	<1ms	—	Eligible amount ₹0 is within the ₹5,000 per-claim limit.
19	fraud_checker	velocity and threshold rules	PASS	<1ms	—	No fraud signals: same-day, monthly, and high-value checks all within limits.
20	decision_synthesizer	confidence rollup	PASS	<1ms	—	Confidence 0.98 = base 0.98; no penalties.
21	decision_synthesizer	final decision	FAIL	<1ms	—	REJECTED; approved amount ₹0; confidence 0.98.